Vision Annotation Audit

Portfolio project | Ahmad Essam Abdelaziz | 7 November 2026

Status: local MVP implemented and verified. GitHub publication blocked: the dedicated repository does not exist and the connected GitHub integration does not expose repository creation. No remote CI or deployment is claimed.

1. What it does

An offline COCO object-detection annotation checker for computer-vision students and dataset curators. It inspects dataset JSON before training to identify broken image/category references, duplicate IDs, invalid bounding boxes, out-of-bounds coordinates, suspicious area mismatches, duplicate boxes and unused categories.

2. Research and ranking

Official 2026/27 employer postings show demand for Python, dataset curation, error analysis, testing and clear communication: Simbe Robotics CV Intern, Zoox Data Science Intern, Amazon ML SDE Intern and Graphcore UK QA Graduate. These are signals of skills demand, not proof of application eligibility.

Rank / concept	Editorial score
1. Vision Annotation Audit	91 / 100
2. SQL Query Plan Explainer	86 / 100
3. Incident Log Correlator	84 / 100
4. Model Inference Profiler	81 / 100

Rubric: hiring demand 25, gap coverage 20, usefulness 15, differentiation 15, feasibility 15, cost 10. Selection avoids repeating the existing Outbox Reliability Lab, TrainServe Sentinel and API Contract Sentinel.

3. Architecture and flow

COCO JSON file → JSON parser → schema / reference / geometry / duplication checks → deterministic issue records → JSON and Markdown reports → CLI exit code.

Core module: **src/vision_audit/core.py** (pure audit function). CLI: **src/vision_audit/__main__.py** (input, reports, status). Fixtures: **examples/**. Tests: **tests/test_core.py**. GitHub Actions workflow: **.github/workflows/ci.yml**. Master instructions: **MASTER_PROMPT.md**.

4. Reproduce the project

```
python -m pip install -e .
python -m vision_audit examples/clean.json --json-out clean.json --md-out clean.md
python -m vision_audit examples/corrupt.json --json-out bad.json --md-out bad.md
python -m unittest discover -s tests -v
```

Exit codes: 0 = no errors, 2 = annotation errors, 3 = input/output error. Warnings alone do not fail the quality gate.

5. Actual test evidence

20/20 unit tests passed; Python compilation passed. Synthetic clean sample: 1 image, 1 annotation, 1 category, 0 errors, 0 warnings, exit 0. Synthetic corrupt sample: 2 image records, 3 annotation records, 2 categories, 8 errors and 3 warnings, exit 2. The corrupt fixture includes duplicate IDs, broken references, invalid geometry, area discrepancy and an unused category. No model accuracy is measured.

6. Boundaries and limitations

No image pixel loading, segmentation/keypoint validation, real-image existence checks, model training or deployed service. COCO area may be segmentation area, so disagreement with bounding-box area is only a warning. Exact matching boxes can be legitimate instances. This lightweight auditor does not replace CVAT, FiftyOne or Datumaro.

7. Interview explanation and truthful CV bullet

Explain: “I created a deterministic data-quality gate for COCO annotation files, with structured error reporting, exit codes for CI and 20 automated tests. I separated validation logic from I/O, used synthetic corrupt fixtures to test failure handling, and documented limitations of heuristic checks.”

CV bullet: Built an offline Python COCO annotation audit CLI with 20 passing tests, deterministic JSON/Markdown reports and CI-ready exit codes; validated duplicate IDs, broken references, invalid bounding boxes and category coverage.

8. GitHub publication checkpoint

Target repository: **Ahmad-Joker/vision-annotation-audit** (not created). Intended branch: **feat/vision-annotation-audit-mvp**. No commit, push, PR or remote CI has been verified. To finish: create the standalone repository with a README, then publish source and PDF, open an unmerged PR against main, and verify CI. The ZIP preserves the complete implementation.

9. Source links

https://jobs.lever.co/SimbeRobotics/d8e862ca-212d-4a88-af74-de400b0c6bf8

https://jobs.lever.co/zoox/85c6aec7-8da2-445d-9c4d-812365dbce97

https://www.amazon.jobs/en/jobs/10593521/software-development-engineer-intern-machine-learning-automated-reasoning-summer-2027

https://job-boards.greenhouse.io/graphcore/jobs/8894574002